

National Physics Olympiad 2015
Group C
30th January, 2015

Water Waves

The behavior of waves on water in shallow channels under the action of gravity is quite complicated but nonetheless can be simplified by using simple pictures.

One can think of the molecules of water undergoing a circular motion while being exposed to a translation motion. The translational motion will coincide with the propagation speed of the water wave.

(a) Express the velocity of a water wave in terms of the wavelength λ , the density of water ρ , its amplitude A and the acceleration due to gravity g . (Hint: You can use the concept of energy conservation). (Hint: Use Energy conservation).

(b) Verify that your result is correct using dimensional analysis.

(3+2)=5

Colliding Photons

Consider a photon of Energy E which hits a (charged) particle of mass M .

a) What will the maximum energy lost by the photon? Consider one dimensional collisions to make your calculations easier. Your result will only contain E, M and some fundamental constants of nature.

b) What will happen if the collision is not head on (i.e. not one dimensional) ? Give arguments in support of your answer. You will not get marks if you just guess even if your answer is right.

Hint: The momentum of massless particles like the photon is given by , $p = \frac{E}{c}$ where c is the speed of light.

(4+1)=5

Liquid Drop Model

The nucleus is bound together due to the the strong nuclear force and one can think of the protons and neutrons inside them (which we collectively call as nucleons) as hard (i.e. solid) incompressible balls. To a good approximation both these types of particles are identical in all respect except that the protons carry electrical charge. The strong nuclear force between nucleons is short-ranged and has saturation properties. This means that each nucleon only has strong nuclear interaction with its neighboring nucleon (irrespective of whether it is a proton or neutron).

a) How should the radius of the nucleus vary as the total number of nucleons (which we will denote by A) ? In other words, if the total number of the nucleons is A and the radius of the nucleus varies as $r \propto A^n$, what is the value of n ?

b) The binding energy is the energy required to break up a system into its individual constituents. The binding energy for a nucleus has contributions from different parts.

i) The bulk effect - This is part that appears from the saturation and short-ranged nature - each nucleon will contribute identically.

ii) The surface effect - Larger nuclei will have larger surface area and thus easier to break. Think of the contribution of the surface tension. The surface tension of liquids arises due to intermolecular forces but inside nucleus the contribution comes from nuclear interaction between nucleons.

iii) Electrostatic repulsion - The electrostatic repulsion between the protons destabilizes the nucleus. The electrostatic repulsion between the protons is long ranged and acts pairwise.

Hence write down a formula for the total binding energy of the nucleus as a function of the nucleon number A and the proton number Z , using three *positive* unknown parameters a_1, a_2, a_3 . Your formula should contain only these five variables and nothing else.

(1+4)=5

Atoms in Solids

1. The harmonic oscillator has a potential energy given by

$$V_0(x) = \frac{1}{2}Kx^2$$

Note that the potential is symmetric under $x \leftrightarrow -x$.

- a) What is the displacement of the oscillator averaged over one period? Why?

In the subsequent parts

- b) The probability of having an energy E is given by

$$p(E) = Ae^{-\beta E}$$

where A is a proportionality constant which is fixed by the requirement that the sum of all probabilities is one, i.e. $\int_0^\infty dE p(E) = 1$ and $\beta = \frac{1}{k_B T}$ with k_B being the Boltzmann constant.

The average value of the a function of energy $f(E)$ hence, is given by

$$\overline{f(E)} \equiv \int_0^\infty p(E)f(E)$$

Hence finds the average value of x^2 .

You will need to use the mathematical identity

$$\int_0^\infty z^n e^{-az} dz = \frac{n!}{a^{n+1}}$$

In reality the molecules in a solid are exposed to a potential

$$V(x) = \frac{1}{2}kx^2 - gx^3$$

- c) Provide a rough sketch of the graph, showing its important features.
- d) Find the expression for the force experienced by the molecules.
- e) In the following- think of the extra gx^3 as a small addition (which we often call as a perturbation) - so that main properties of the system do not drastically differ from the system in the absence of the interactions.
Each of the atoms in solids can be thought of as a particle with a particular form of potential energy. Find the average equilibrium position of the atoms as a function of temperature.
- f) What is the significance of your result?

(1+2+2+3+2)=10

What do you make of it?

Consider an experiment where a semi-transparent glass with a thickness d is illuminated with a light of intensity I_0 and the intensity I of the transmitted light is measured behind the glass. The formula between d and I is given by

$$I(d) = I_0 \exp\left(-\frac{d}{\lambda}\right) = I_0 e^{-\frac{d}{\lambda}}$$

where the constant λ is called the extinction length in the glass. The intensity of I is measured using glass of different thicknesses but of the same material.

The data is shown in the table

d	I
2.5 ± 0.5	4.34 ± 0.1
6.0	1.89
8.0	0.822 ± 0.005
10.5	0.365
12.0	0.195

Find the value of λ along with the error in it. Give an estimation of the error

(3.5+1.5)=5